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# Paleomagnetism, paleogeographic origins, and uplift history of the Coast Range ophiolite at Mount Diablo, California

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**Abstract.** Divergent paleogeographic origins have been proposed for the Coast Range ophiolite of western California which are testable using paleomagnetic methods. Paleomagnetic data for Middle Jurassic pillow lavas and diabase sills of the Coast Range ophiolite at Mount Diablo, northern California, indicate that they contain two components of remanent magnetization. The characteristic directions have normal and reversed polarities and apparently are carried by Ti-poor magnetite. This magnetization is inferred to have been acquired during emplacement and seafloor alteration at an ancient spreading ridge. The paleolatitude calculated from its structurally corrected mean direction is 20°N ± 9° and agrees with the expected direction for stable North America; this result is also consistent with the concordant paleolatitude (32°N ± 8°) recently determined for Upper Jurassic Coast Range ophiolite at Stanley Mountain in southern California. In addition, clockwise vertical axis rotation of Mount Diablo (143° ± 11°) is indicated by the characteristic magnetization direction. An overprint component is inferred to have been acquired during uplift of Mount Diablo since the Miocene.

## 1. Introduction

Three contrasting models for genesis of Jurassic ophiolitic rocks in the western Coast Ranges of California have recently been discussed by *Dickinson et al.* [1996]. Although these models share broad similarities within a plate tectonic framework, each hypothesis has distinct implications for development of the continental margin. Points of origin outlined for the Coast Range ophiolite are (1) backarc spreading behind an east facing intraoceanic subduction zone, (2) mid-ocean spreading at equatorial paleolatitudes, and (3) spreading within the forearc region due to transtensional deformation. While the first and third hypotheses require little or no latitudinal displacements relative to the North American craton, the second implies significant northward translation from the paleoequator. Paleomagnetic data therefore would be useful in evaluating these alternatives.

Reliable paleomagnetic results bearing on the paleogeographic origins of the Coast Range ophiolite, however, have been difficult to obtain. In northern California, characteristic remanent magnetizations for samples from the Del Puerto Canyon, Stonyford, and Paskenta regions postdate folding, show only normal polarity where both polarities are expected, and tend to have directions similar to younger geomagnetic field directions. These magnetizations are clearly secondary and were acquired during either deep subaerial weathering, burial and uplift, or low-temperature chemical alteration of these ophiolitic rocks [*Frei and Blake*, 1987]. Limited data from two other localities (Mount Diablo, Potter Valley) show paleolatitudes concordant with Jurassic North America [*Mankinen et al.*, 1991], but fold and reversal tests were unavailable.

In southern California, characteristic magnetizations for remnants of the Coast Range ophiolite (Stanley Mountain, Point Sal, Cuesta Ridge, Llanada) have been consistently interpreted as having been acquired at low paleolatitudes [*Luyendyk and Hornafius*, 1982; *McWilliams and Howell*, 1982; *Beebe and Luyendyk*, 1983], but field stability tests are also lacking, and only normal polarity is indicated at each locality; therefore these results are also consistent with subsequent remagnetization. Recent studies of ophiolitic rocks at Stanley Mountain [*Hagstrum*, 1992; *Hagstrum and Murchey*, 1996] and on the Vizcaino Peninsula in Baja California [*Hagstrum et al.*, 1993] indicate definitively that these rocks have been magnetically overprinted.

Similarly, oceanic rocks of the Franciscan Complex have been largely remagnetized, but high-unblocking temperature components of magnetization showing normal and reversed polarities, consistent structurally corrected results, and deposition at equatorial paleolatitudes, have been found in red radiolarian chert sequences at six localities in California and Mexico [*Hagstrum and Sedlock*, 1992; *Hagstrum and Murchey*, 1993; *Hagstrum et al.*, 1996]. Original magnetizations may be irregularly preserved within some parts of the Coast Range ophiolite as well, as has been inferred for red tuffaceous chert at Stanley Mountain [*Hagstrum and Murchey*, 1996].

This investigation was undertaken to determine if pillow basalt and diabasic rocks of the Coast Range ophiolite at Mount Diablo also retain an original magnetization. Although the diabasic rocks were the subject of a previous paleo-

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magnetic investigation by *Mankinen et al.* [1991], we have returned to the area for two reasons: first, *Mankinen et al.* [1991] were unsuccessful in isolating a characteristic component of magnetization that could be fitted with least squares lines, and second, subsequent geologic mapping indicates that what they interpreted to be sheeted dikes are most likely sheeted sills.

## 2. Geologic Setting and Sampling

Mount Diablo, the highest peak in the eastern part of the central Coast Ranges, is situated at the north end of the Diablo Range in Contra Costa County and forms the structural culmination of the Altamont anticline [*Jones et al.*, 1994]. The remnant of Coast Range ophiolite exposed at Mount Diablo (Figure 1) represents a fragment of mafic to ultramafic oceanic crust and is Middle to Late Jurassic in age; zircons from leucocratic dikelets within it have a U-Pb age of 165 Ma (J. M. Mattinson as given by *Mankinen et al.* [1991]). Additionally, *Mattinson and Hopson* [1992] point out that U-Pb zircon ages from the Coast Range ophiolite tend to be too young by ~5 to 10 Ma. The Coast Range ophiolite, in general, is depositionally overlain by the Great Valley Group [*Bailey et al.*, 1970], although the contact between them is a fault everywhere in Contra Costa County [*Graymer et al.*, 1994]. Abundant fossils of the bivalve *Buchia* are locally contained in the lower part of the Great Valley Group, the Late Jurassic to Early Cretaceous Knoxville Formation. Sheared sedimentary and metamorphic rocks with blocks and zones of basalt and chert, and mélange, make up the Franciscan Complex which forms the bulk of Mount Diablo. Franciscan Complex rocks were emplaced below the Coast Range ophiolite by subduction and accretionary faulting during Cretaceous time (the contact now represented by the Coast Range fault), and presumably underlie all of Contra Costa County [*Graymer et al.*, 1994].

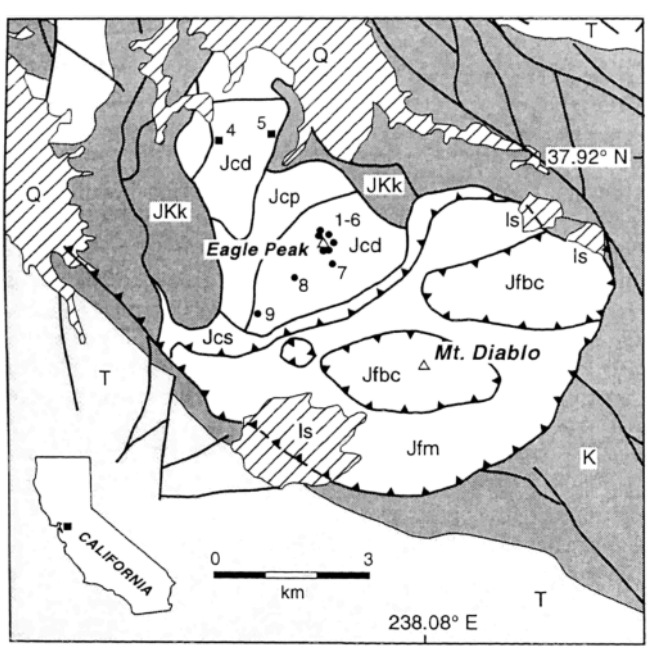



**Figure 1.** Geologic map of the Mount Diablo area, northern California [after *Graymer et al.*, 1994], showing locations of the paleomagnetic sampling sites: circles (1-9), this study; squares (4, 5), sites of *Mankinen et al.* [1991]. Jcd, Jurassic Coast Range ophiolite (CRO) diabase: mainly sills with some dikes, including screens of pillow basalt; Jcp, Jurassic CRO pillow basalt; Jcs, Jurassic CRO serpentinite; Jfbc, Jurassic Franciscan Complex pillow basalt and radiolarian chert; Jfm, Jurassic Franciscan Complex mélange; JKk, Late Jurassic to Early Cretaceous Knoxville Formation; K, Cretaceous rocks undivided; T, Tertiary rocks undivided; Q, Quaternary surficial and landslide (ls) deposits.

Between 6 and 13 samples of the Coast Range ophiolite were collected for paleomagnetic analysis at each of nine sites in the freshest exposures available on the northern flank of Mount Diablo (Figure 1). Sampling included pillow basalts and two diabase dikes at site MD1, pillow basalts and a sill at site MD4, diabase and kyratophyre dikes at MD2, pillow basalts only at sites MD3, MD5, MD7, and MD9, and sheeted sills at sites MD6 and MD8. A portable, battery-powered gradiometer was used to help avoid lightning-struck outcrops. Sites MD1 to MD7 near Eagle Peak and site MD9 on Olofson Ridge are at elevations between 610 and 730 m. Screens of pillow basalt within the sheeted sill complex on Eagle Peak are more common at higher elevations, and upper crowns and lower keels in the pillow forms provide an estimate of paleohorizontal which is subparallel to the sheeted sill contacts. The regional attitude (strike/dip) based on a large number of measurements in the Eagle Peak area is 195°/35°W, although the structural attitude at site MD9 is 45°/50°E. Site MD8 is in the streambed of Mitchell Creek at the bottom of Mitchell Canyon (elevation of 290 m), and no pillow basalts were found there. Although the stream-wash exposures appeared unweathered, sulfide minerals (pyrite?) are apparent in the paleomagnetic samples of these sills.

*Mankinen et al.* [1991] sampled sheeted intrusives in the Kaiser and Lone Star quarries (sites 4 and 5, Figure 1) at a similar elevation to our site in Mitchell Creek (MD8); *Williams* [1983] also noted pyrite in microcrystalline diabase "dikes" exposed in the quarries. No pillow screens or other paleohorizontal indicator is available in the quarries, so *Mankinen et al.* [1991] used the attitude of pillow basalts at Olofson Ridge, cut by a diabase dike subparallel to the quarry intrusives, to correct their paleomagnetic data. However, the similar attitude and identical appearance of the sheeted intrusives sampled by *Mankinen et al.* [1991] to the sheeted sills we sampled near Eagle Peak indicate that the rocks in the Kaiser and Lone Star quarries are also sheeted sills.

## 3. Laboratory Methods and Results

The remanent magnetization of individual specimens, cut from the core samples, was measured with a cryogenic magnetometer. Specimens from all samples were progressively demagnetized with either standard alternating field (af) or thermal procedures to 100 mT or 630°C, respectively. Two components of magnetization were found in these rocks, and lines representing the secondary and characteristic magnetization directions were fitted to the demagnetization data using a least squares method based on principal component analysis [*Kirschvink*, 1980]. The statistics of *Fisher* [1953] are used in analyzing the site and overall mean directions for the secondary component, and Bingham statistics [*Onstott*, 1980] are used in analyzing the streaked mean directions of the characteristic component (Table 1).

Figure 2 shows representative demagnetization data (in situ) for samples of the Coast Range ophiolite at Mount Diablo in

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**Table 1.** Paleomagnetic Data for the Coast Range Ophiolite at Mount Diablo, California

| Sites | $I$ | $D$ | $N/N_0$ | $R$ | $k$ | $-k_1/-k_2$ | $\alpha_{95}$ | $\alpha_1/\alpha_2$ | $\lambda$ | $\phi$ |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | *Overprint Component* | | | | | |
| MD2 | 64.5 | 17.1 | 3/9 | 2.9223 | 26 | | 24.8 | | 74.9 | 289.0 |
| MD3 | 60.4 | 355.1 | 8/8 | 7.9073 | 76 | | 6.4 | | 84.9 | 191.8 |
| MD4 | 59.1 | 352.6 | 7/13 | 6.8639 | 44 | | 9.2 | | 83.9 | 169.7 |
| MD5 | 61.1 | 9.8 | 5/8 | 4.9585 | 96 | | 7.8 | | 81.4 | 295.7 |
| MD6 | 58.6 | 8.4 | 6/7 | 5.9076 | 54 | | 9.2 | | 83.3 | 313.4 |
| MD7 | 66.8 | 349.2 | 4/8 | 3.9355 | 46 | | 13.6 | | 76.1 | 207.7 |
| MD8 | 72.4 | 353.4 | 6/6 | 5.9714 | 175 | | 5.1 | | 69.8 | 227.8 |
| MD9 | 65.0 | 352.4 | 8/10 | 7.8592 | 50 | | 7.9 | | 79.4 | 208.9 |
| Avg1 | 63.8 | 0.2 | (8/9) | 7.9545 | 154 | | 4.5 | | | |
| | | | (8/9) | 7.9043 | 73 | | 6.5 | | 82.1 | 238.4 |
| Avg2 | 63.4 | 358.6 | 47 | 46.0833 | 50 | | 3.0 | | 82.8 | 230.3 |
| | | | | | *Characteristic Component* | | | | | |
| MD1 | -43.4 | 316.6 | 11/12 | | | -13/-3 | | 10.1/25.7 | 14.8 | 98.1 |
| MD2 | 49.5 | 152.7 | 3/9 | | | -701/-14 | | 2.2/15.6 | 17.2 | 262.5 |
| MD4 | -23.8 | 311.4 | 9/13 | | | -12/-8 | | 10.4/13.3 | 22.2 | 110.4 |
| MD5 | 12.4 | 122.2 | 5/8 | | | -17/-4 | | 12.0/28.8 | -20.5 | 302.0 |
| MD6 | -60.6 | 290.5 | 2/7 | | | | | | -11.6 | 103.8 |
| MD7 | -12.8 | 306.3 | 3/8 | | | -37/-2 | | 10.8/48.5 | 23.2 | 118.8 |
| MD8 | 49.9 | 120.7 | 2/6 | | | | | | -1.9 | 285.8 |
| Avg1 | 36.7 | 129.0 | (7/9) | | | -28/-5 | | 7.6/18.8 | | |
| | | | (7/9) | | | -43/-5 | | 6.1/18.6 | -8.5 | 286.2 |
| Avg2 | -33.8 | 311.3 | 35 | | | -7/-4 | | 7.8/11.4 | 17.4 | 106.4 |
| AvgN | 39.3 | 124.2 | 16 | | | -10/-3 | | 9.5/18.2 | -10.3 | 289.2 |
| AvgR | -29.4 | 317.2 | 19 | | | -9/-4 | | 8.7/15.4 | 23.0 | 103.4 |

$I$ and $D$ are inclination and declination of paleomagnetic directions in degrees; overprint directions uncorrected for tilt, characteristic directions corrected for regional structural tilt (strike is 195°, dip is 35°W). $N/N_0$ is number of samples (sites) analyzed/collected. For uncorrected mean directions, $R$ is vector sum of $N$ unit vectors; $k$ is the concentration parameter [*Fisher*, 1953]; and $\alpha_{95}$ is the radius of 95% confidence in degrees. For corrected mean directions, $R$ is not calculated, $-k_1/-k_2$ are the two Bingham concentration parameters, and $\alpha_1/\alpha_2$ are the two Bingham 95% confidence limits [*Onstott*, 1980]. $\lambda$ and $\phi$ are latitude and east longitude, respectively, of virtual geomagnetic poles, in degrees. For Avg 1, statistics for averaged poles are also given. Sampling locality is 37.9°N, 238.1°E.

which both the characteristic and secondary components of magnetization could be fitted with least squares lines. In most cases, lines were fitted to only one of the two components within the data set for a given specimen; the demagnetization trajectories are most often univectorial (Figures 2b and 2c). At site MD1, least squares lines could not be fitted to the overprint component, although its presence can be inferred from the lower demagnetization steps of some samples (Figure 2a); both the characteristic and overprint components could be isolated in sample MD4-3 (Figure 2d). In general, complete demagnetization of the remanent magnetization was accomplished by thermal heatings to ~580°C, indicating that the magnetic carrier in these rocks is magnetite or Ti-poor titanomagnetite. In some samples (Figure 2c), there also appears to be a small portion of a higher unblocking temperature mineral like hematite.

In samples from a number of sites, neither the characteristic direction nor the overprint direction could be isolated. Demagnetization data for such a sample are shown in Figure 3. Two components are apparent in the orthogonal vector plot (Figure 3a), but compared to the characteristic and overprint directions their inclinations are too steep; the data points are in between the overprint and characteristic directions (Figure 3b). This is the problem *Mankinen et al.* [1991] faced with their samples from the sheeted sills in the quarries: an inability to completely separate the two components during demagnetization. We may not have succeeded entirely in this study either, and the two components may still be influenced by one another causing the overprint direction to be too steep and the characteristic directions to be streaked out. Sample directions are plotted in Figure 4, and site-level (Avg1) and sample-level (Avg2) overall means are given in Table 1.

The in situ secondary component is of normal polarity (Figure 4) with moderate to steep inclinations and northward declinations (Table 1). The mean direction (Avg1: $I = 63.8°$, $D = 0.2°$) is similar to, but statistically different from (95% confidence), both the expected present-day ($I = 61.9°$, $D = 16.0°$) and Miocene ($I = 56.5°$, $D = 356.9°$ [see *Hagstrum et al.*, 1987]) field directions at Mount Diablo. The overprint direction is also indistinguishable from an expected Late Cretaceous direction for Mount Diablo ($I = 63.2°$, $D = 354.8°$) determined using a reference pole from the Adel Mountains of Montana [*Gunderson and Sheriff*, 1991]. The characteristic component, on the other hand, shows two polarities and passes a reversal test at the 95% confidence level with a "C" classification ($\gamma_c = 14.6°$) [*McFadden and McElhinny*, 1990]. Our in situ characteristic direction (Avg1: $I = 4.0°$, $D = 124.1°$) is similar to the stable endpoint direction ($I = 7.8°$, $D = 119.2°$) reported by *Mankinen et al.* [1991]. This similarity further supports our interpretation that the sheeted intrusives in the Kaiser and Lone Star quarries are sills.

A paleolatitude calculated from our structurally corrected

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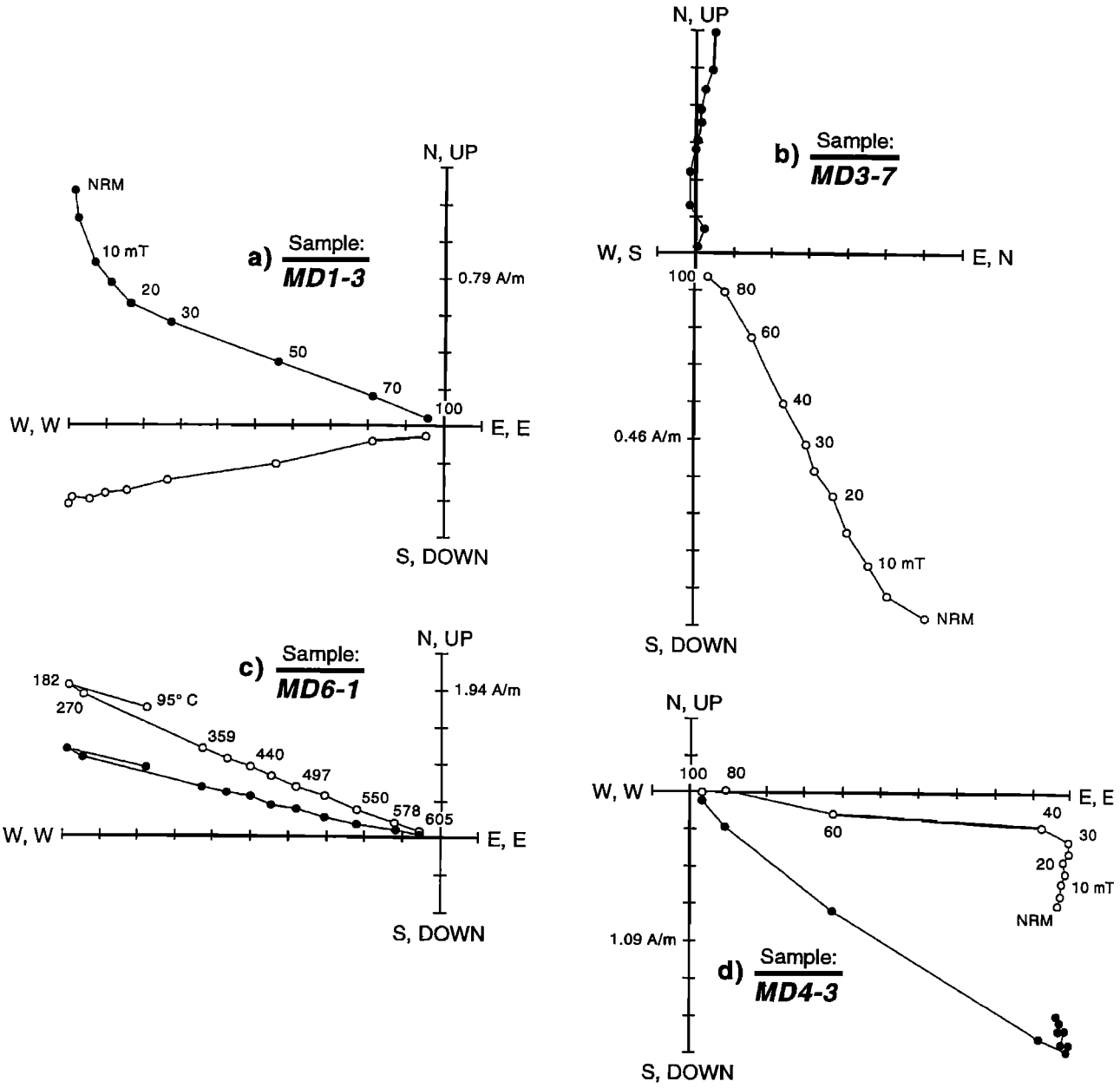



**Figure 2.** Orthogonal projections (in situ) of stepwise alternating field (mT) and thermal (°C) demagnetization curves for samples of the Coast Range ophiolite at Mount Diablo. (a) Sample MD1-3 of diabase dike showing characteristic component and vestige of overprint component at lower demagnetization steps; (b) sample MD3-7 of pillow basalt showing only overprint component; (c) sample MD6-1 of sills showing only characteristic component; and (d) sample MD4-3 of pillow basalt showing both characteristic and secondary components of remanent magnetization. Solid (open) symbols indicate projection onto the horizontal (vertical) plane.

characteristic component (Table 1, Avg1) is 20° ± 9°. Coincidentally, a similar estimate of paleolatitude (27° ± 4°) is calculated from the structurally corrected direction reported by *Mankinen et al.* [1991] (assuming sheeted dikes), and they inferred the corresponding vertical axis rotation to be 53° ± 5° counterclockwise. We, however, determined a clockwise rotation of 143° ± 11° relative to stable North America; both of these rotations [*Demarest*, 1983] are based on a Middle Jurassic reference pole (67°N, 105°E, $A_{95}$ = 3°) for North America calculated from African, Eurasian, North American, and Indian plate reference poles pooled in a common reference frame [*Besse and Courtillot*, 1991]. Clockwise vertical axis rotation of Mount Diablo (~140°) agrees with that inferred for the nearby Marin Headlands terrane (~130°) based on geologic and paleomagnetic evidence [*Wahrhaftig*, 1984; *Curry et al.*, 1984; *Hagstrum and Murchey*, 1993].

## 4. Discussion and Conclusions

Samples of the diabase dikes at site MD1 have "normal" polarity, whereas their host pillow basalts have "reversed"

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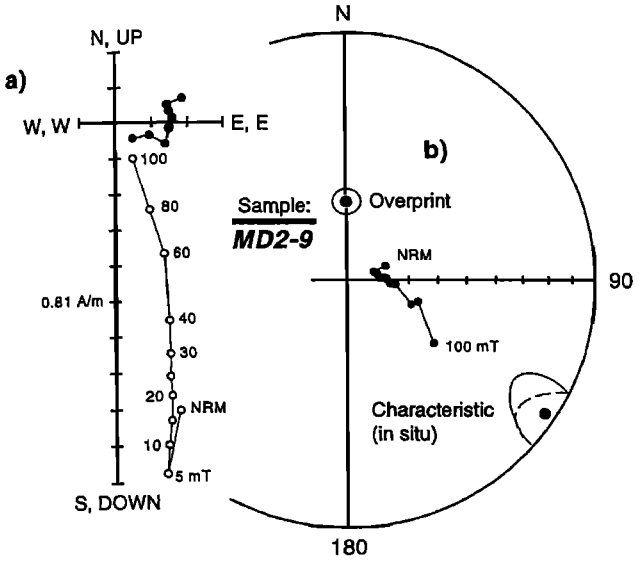



**Figure 3.** Orthogonal and equal-area projection of in situ demagnetization data for sample MD2-9 of diabase dike. Vestiges of the characteristic and overprint components are apparent but cannot be separated with the demagnetization treatments (alternating field). Data points fall between the in situ overprint and characteristic directions (with 95% confidence limits; dashed in upper hemisphere) and have been discarded from the overall means.

polarity characteristic magnetizations (Figure 4 and Table 1), implying that these rocks retain their original thermoremanent magnetizations. Also, pillow basalts and sills at sites MD4 through MD7 each have both normal and reversed polarity magnetizations within individual sites ($<10^2$ m). These magnetizations may be original thermoremanent magnetizations but also could be thermochemical remagnetizations related to subsequent dike and sill emplacement and/or hydrothermal alteration. The Coast Range ophiolite at Mount Diablo has been subjected to hydrothermal alteration at a spreading ridge [*Williams*, 1983], and acquisition of the characteristic component of magnetization likely occurred, at least in part, during that time. A positive reversal test indicates that the characteristic magnetization was acquired spanning at least two polarity intervals of the geomagnetic field, but no fold test is available to help constrain the timing of this event; a consistent characteristic magnetization was not found at site MD9 on Olofson Ridge.

The secondary component is of normal polarity and has a direction similar to that of the Late Cretaceous through late Cenozoic dipole field for North America. Although this secondary direction is most similar to the expected Late Cretaceous direction, the Coast Range ophiolite at Mount Diablo was part of the upper plate and not subjected to low-temperature, high-pressure alteration associated with subduction along the California margin. Also, the secondary component might have steeper inclinations due to overlap with the characteristic component. Therefore the secondary component is inferred to be Miocene or younger in age and is possibly related to deep burial beneath the Great Valley Group and recent uplift of Mount Diablo and the Altamont anticline [*Pullaiah et al.*, 1975]. Because secondary components of magnetization with similar directions have been found in Coast Range ophiolite and Franciscan Complex rocks at a number of sites along the western margin of North America (J. T. Hagstrum and R. L. Sedlock, Paleomagnetism of Cretaceous forearc strata on Santa Margarita and Magdalena Islands, Baja California Sur: Implications for uplift and remagnetization along the California margin, submitted to *Tectonics*, 1997), we prefer this explanation for the origin of the secondary component. Presence of the Lawlor Tuff south of Mount Diablo implies that this topographic feature did not exist at the time of eruption (4.0 to 4.5 Ma) and that it probably formed later [*Sarna-Wojcicki*, 1972]. Furthermore, uplift within the Coast Ranges of California might also be related to increased compression between the Pacific and North American plates that occurred from 3.9 to 3.4 Ma [*Harbert*, 1991]. Another possibility, however, is that the secondary component is related to late Miocene (rhyolite stocks, dikes, and sills) or Pliocene (basalt plugs and dikes) igneous activity in the region, although none of these outcrop on Mount Diablo [see *Graymer et al.*, 1994].

Without knowing the polarity of the geomagnetic field at the time of magnetization acquisition in the Jurassic, it is permissible that the characteristic component was acquired in either the northern or southern hemisphere near 20° paleolatitude. Formation of the Coast Range ophiolite in the northern hemisphere is clearly preferred, however, because there are no known plate motions that could have transported it from 20°S within the alotted time frame [*Engebretson et al.*, 1985]. The available Middle Jurassic paleomagnetic poles for North America give conflicting results, so the position of the North American plate at that time is poorly defined [*Hagstrum*,

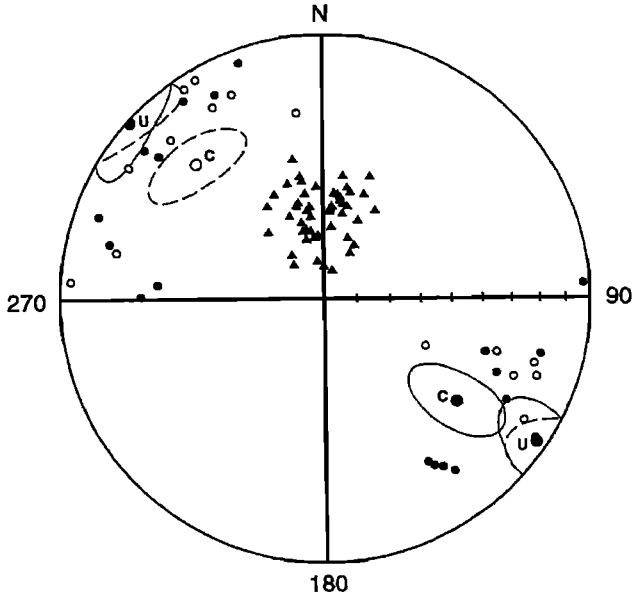



**Figure 4.** Equal-area projection of paleomagnetic directions for the Coast Range ophiolite at Mount Diablo. Triangles and small circles indicate in situ sample-mean directions for the overprint and characteristic components of magnetization, respectively. Large circles with 95% confidence limits (dashed in the upper hemisphere) indicate mean normal and reversed polarity directions of the characteristic component uncorrected (U) and corrected (C) for structural tilt. Solid (open) symbols indicate projection onto the lower (upper) hemisphere.

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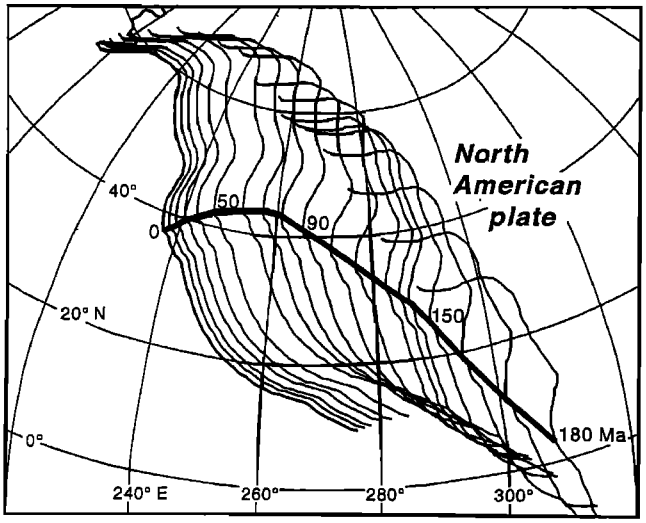



**Figure 5.** Motion of the North American plate relative to the hot spots for the past 180 Myr [after Figure 10a of *Engebretson et al.*, 1985]. Shown is the western coastline of present-day North America at 10 m.y. intervals; thick line is the trajectory of a point (San Francisco) near Mount Diablo's present location.

1993]. Perhaps the best reference poles available for the North American plate during the Jurassic are those synthesized from reference poles for a number of plates (African, Eurasian, North American, Indian) by *Besse and Courtillot* [1991].

A comparison of the 170 Ma reference pole [*Besse and Courtillot*, 1991] with the observed pole for Mount Diablo (≥165 Ma) indicates a latitude shift of -5° ± 10° (southward) relative to North America. A comparison of the 150 Ma reference pole [*Besse and Courtillot*, 1991] with the observed pole for Upper Jurassic radiolarian chert above the Coast Range ophiolite at Stanley Mountain in southern California [*Hagstrum and Murchey*, 1996] indicates an latitude shift of 1° ± 8° (northward) relative to North America. Thus the paleomagnetic data available for the Coast Range ophiolite at Mount Diablo and at Stanley Mountain do not indicate significant transport of these rocks with respect to the North American craton. However, the possibility of southward displacement, followed by northward displacement, of these terranes exists [*Kelley and Engebretson*, 1994]. Figure 5 shows the general movement of the North American plate relative to the hot spots for the past 180 Myr [*Engebretson et al.*, 1985]. North America moved rapidly northward in the Jurassic, and a 20°N paleolatitude would be expected for the older rocks at Mount Diablo, compared to the 32°N paleolatitude for younger rocks at Stanley Mountain to the south.

Origin of the Coast Range ophiolite at Mount Diablo near the paleoequator, similar to a model proposed for ophiolite fragments in southern California [*Hopson et al.*, 1997], is not supported by our data. Moreover, the characteristic remanent magnetization of the pillow basalts at Stanley Mountain, indicating equatorial paleolatitudes [*McWilliams and Howell*, 1982], has been shown to be a later overprint magnetization [*Hagstrum*, 1992; *Hagstrum and Murchey*, 1996]. In addition, serious problems exist with the paleobiogeographic models (based on radiolarian distribution patterns) which also have been used to support equatorial origins for the southern Coast Range ophiolite [*Hopson et al.*, 1997]. These problems include the schematic nature of the models and their lack of both quantitative population data and paleomagnetic data, their reliance on faunal data from allochthonous terranes, and the possibility of environmental controls other than latitude on the radiolarian distribution patterns [*Murchey and Hagstrum*, 1997].

An origin of the Coast Range ophiolite near the paleoequator would require rapid northward movement of an unknown plate during the Middle to Late Jurassic similar to, but prior to formation of, the Kula plate [*Engebretson et al.*, 1985]. Paleomagnetic results for radiolarian chert of the Franciscan Complex are consistent with an eastward progression of the Farallon plate at this time beneath the equatorial zone of high biologic productivity [*Hagstrum and Murchey*, 1993]. Accretion of the Franciscan chert in southern Mexico in the Cenomanian and subsequent northward translation, likely by transform motion associated with the Kula plate [*Hagstrum and Murchey*, 1993], indicate a substantially different tectonic transport history for this terrane than for the Coast Range ophiolite. These rocks are presently juxtaposed in the San Francisco Bay area, implying that a major tectonic boundary lies between the Marin Headlands and Mount Diablo. Similar clockwise rotations for these two terranes, however, might indicate a common tectonic history postjuxtaposition in the Tertiary, consistent with the overall pattern of dextral shear observed along the western edge of North America [*Beck*, 1980].

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